Set 24: Product Inventory Management

DSA – Data Handling and Visualization | Lab Experiment

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Aim

To analyse product inventory using a bar chart, a stacked bar chart by category and a price-quantity scatter plot in R, and a Tableau dashboard combining the bar chart and stacked bar chart.

Software / Tools Used

R 4.3.3 with ggplot2 (charts), gridExtra (tables) and patchwork (dashboard layout); Tableau Desktop / Tableau Public (interactive dashboard).

Dataset

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

Assumed categories and prices (not given in this dataset):

Product Name	Category (assumed)	Price in $ (assumed)
Product A	Category 1	20
Product B	Category 2	15
Product C	Category 1	18
Product D	Category 2	22
Product E	Category 3	17

Assumption: The dataset has no category or price columns. The categories were assumed for Q2 and Q3. For Q4, prices for Products A–C are taken from Set 18 (20, 15, 18), and the prices of Products D and E (22 and 17) are assumed. Replace them with real values if available.

Program – Dataset Creation (common to all R tasks)

```
library(ggplot2)
library(gridExtra)
library(patchwork)
options(scipen = 999)
# Plain white table theme with black borders (no colours)
tbl_theme <- ttheme_minimal(
  core    = list(fg_params = list(col = "black", fontsize = 11),
                 bg_params = list(fill = "white", col = "black")),
  colhead = list(fg_params = list(col = "black", fontface = "bold", fontsize = 11.5),
                 bg_params = list(fill = "white", col = "black")))
inv <- data.frame(
  Product_ID   = 1:5,
  Product_Name = paste("Product", LETTERS[1:5]),
  Quantity     = c(250, 175, 300, 200, 220),
```

```
  # ASSUMED: categories and prices are not in the dataset
  # (Prices for A-C taken from Set 18; D and E assumed)
  Category     = c("Category 1", "Category 2", "Category 1", "Category 2", "Category 3"),
  Price        = c(20, 15, 18, 22, 17)
)
write.csv(inv, "Product_Inventory.csv", row.names = FALSE)  # for Tableau
pal <- c("Product A" = "#0072B2", "Product B" = "#E69F00", "Product C" = "#009E73",
         "Product D" = "#CC79A7", "Product E" = "#D55E00")
```

Q1. Bar Chart – Quantity of Each Product

> *In R, create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.*

R Program:

```
print(inv[, c("Product_Name", "Quantity")], row.names = FALSE)
cat("Total units:", sum(inv$Quantity), "| Average:", mean(inv$Quantity), "\n")
p_bar <- ggplot(inv, aes(x = Product_Name, y = Quantity, fill = Product_Name)) +
  geom_col(width = 0.6) + geom_text(aes(label = Quantity), vjust = -0.5, size = 4.3) +
  scale_fill_manual(values = pal) + scale_y_continuous(limits = c(0, 350)) +
  labs(title = "Quantity of Each Product in Inventory", x = "Product", y = "Quantity
Available") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5),
legend.position = "none")
print(p_bar)
ggsave("s24_q1.png", p_bar, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
 Product_Name Quantity
    Product A      250
    Product B      175
    Product C      300
    Product D      200
    Product E      220
Total units: 1145 | Average: 229
```

Output:

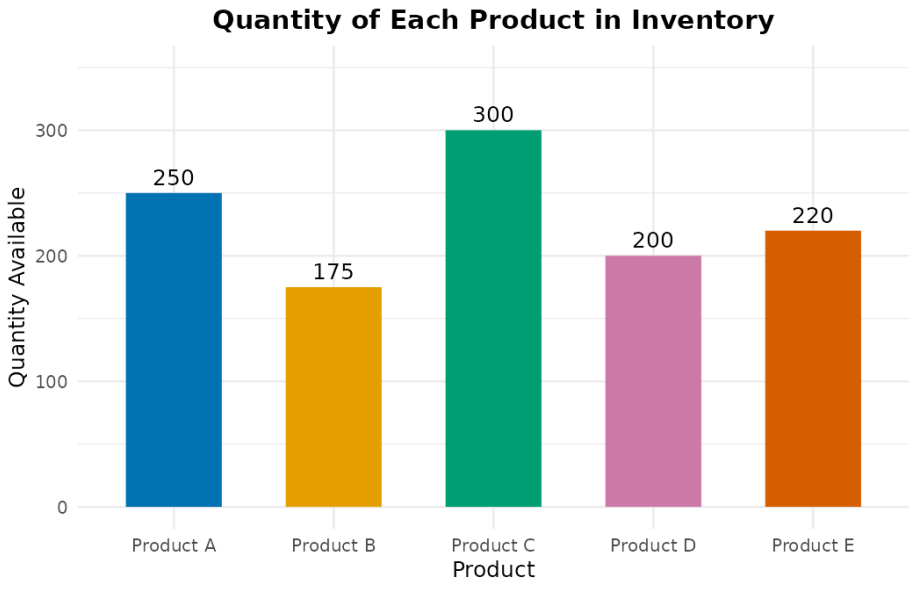


Figure 1: Quantity available per product

Inference:

Product C has the most stock (300 units) and Product B the least (175). Total inventory is 1,145 units, an average of 229 per product.

Q2. Stacked Bar Chart – Quantity within Product Categories

> *In R, generate a stacked bar chart to show the quantity of each product within different product categories.*

R Program:

```
print(xtabs(Quantity ~ Category + Product_Name, data = inv))
p_stk <- ggplot(inv, aes(x = Category, y = Quantity, fill = Product_Name)) +
  geom_col(width = 0.55, colour = "white") +
  geom_text(aes(label = paste0(Product_Name, ": ", Quantity)), position = position_stack(vjust
= 0.5),
            colour = "white", size = 3.5, fontface = "bold") +
  scale_fill_manual(values = pal) +
  labs(title = "Quantity of Each Product within Product Categories", x = "Product Category", y
= "Quantity Available", fill = "Product") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5))
print(p_stk)
ggsave("s24_q2.png", p_stk, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
            Product_Name
Category     Product A Product B Product C Product D Product E
  Category 1       250         0       300         0         0
  Category 2         0       175         0       200         0
  Category 3         0         0         0         0       220
```

Output:

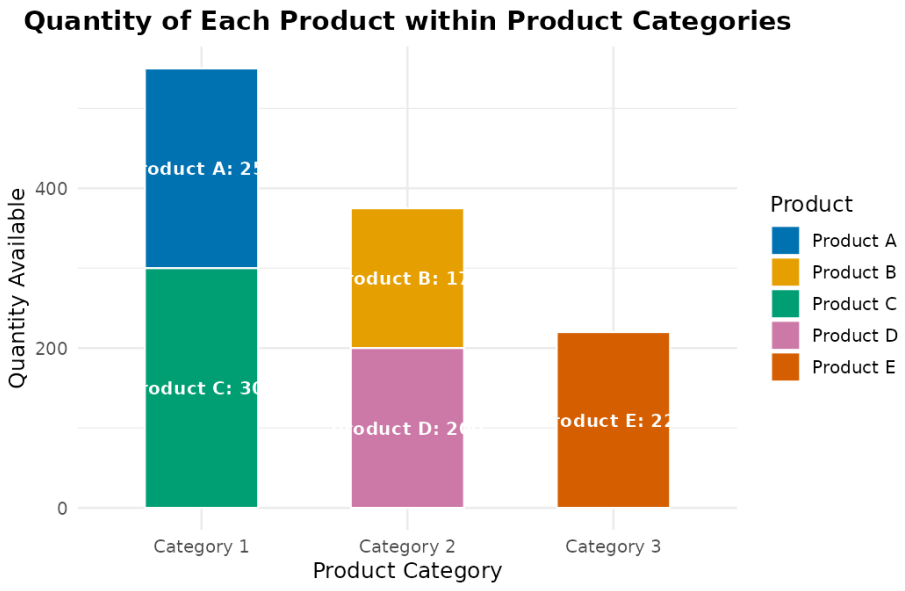


Figure 2: Quantity of each product within assumed categories

Inference:

Category 1 (Products A and C) holds 550 units (48.0%), Category 2 (Products B and D) holds 375 units (32.8%) and Category 3 (Product E) holds 220 units (19.2%). Category 1 carries the largest share of stock.

Q3. Tableau Dashboard – Bar Chart and Stacked Bar Chart

> *In Tableau, build a dashboard combining the bar chart and stacked bar chart for interactive exploration of inventory data.*

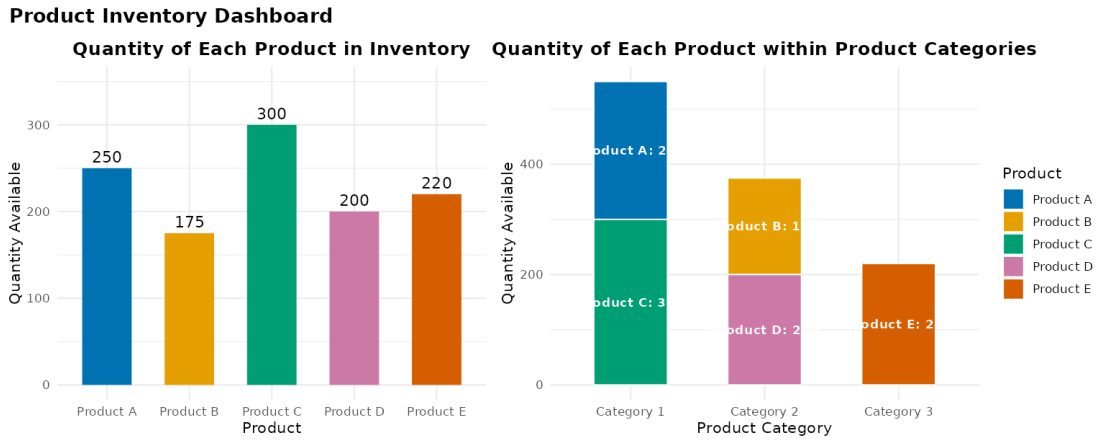
R Program:

```
dash <- (p_bar | p_stk) +
  plot_annotation(title = "Product Inventory Dashboard", theme = theme(plot.title =
element_text(face = "bold", size = 15)))
ggsave("s24_dash.png", dash, width = 12, height = 4.8, dpi = 130, bg = "white")
```

Procedure in Tableau:

1. Open Tableau → **Connect → Text file** → **Product_Inventory.csv**.
2. **Sheet 1 – "Quantity by Product":** drag Product Name to Columns and SUM(Quantity) to Rows. Drag Product Name to Color and show labels. Sort descending by quantity.
3. **Sheet 2 – "Stacked by Category":** drag Category to Columns and SUM(Quantity) to Rows, then drag Product Name to **Color** and Label so each bar is stacked by product.
4. From the menu choose **Dashboard → New Dashboard** and set **Size → Automatic** (or Fixed 1200 × 800).
5. Drag the sheets onto the canvas: Quantity by Product (left) and Stacked by Category (right). Add a **Text** object at the top as the dashboard title.
6. On each chart, open the sheet's drop-down and select **Use as Filter**, so clicking a bar, point or slice filters the other views.
7. Go to **Dashboard → Actions → Add Action → Highlight** (Run on: Hover) so related marks highlight across all views.
8. Right-click each quick filter → **Apply to Worksheets → All Using This Data Source** (Product_Inventory) so one filter controls every view.
9. Add a **Category** quick filter to the dashboard.
10. Save as a packaged workbook (**File → Save As → .twbx**) or publish to Tableau Public.

Dashboard Layout:



Reference layout of the dashboard (drawn in R). The interactive version is built in Tableau using the procedure above.

Inference:

Selecting a category bar highlights the matching products in the bar chart, so quantities can be compared at both product level and category level.

Q4. Scatter Plot – Price vs Quantity Available

> *In R, create a scatter plot to explore the relationship between product price and quantity available. Explain any insights.*

R Program:

```
cat("Correlation (Price vs Quantity):", round(cor(inv$Price, inv$Quantity), 3), "\n")
fit <- lm(Quantity ~ Price, data = inv)
cat("Regression: Quantity =", round(coef(fit)[1], 1), "+", round(coef(fit)[2], 2), "* Price
(R-squared =", round(summary(fit)$r.squared, 3), ")\n")
p_sc <- ggplot(inv, aes(x = Price, y = Quantity)) +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE, colour = "grey50", linetype =
"dashed") +
  geom_point(aes(colour = Product_Name), size = 5) +
  geom_text(aes(label = Product_Name), vjust = -1.4, size = 3.6) +
  scale_colour_manual(values = pal) + scale_x_continuous(limits = c(13, 24)) +
scale_y_continuous(limits = c(150, 340)) +
  labs(title = "Product Price vs Quantity Available (D, E prices assumed)", x = "Price (in
$)", y = "Quantity Available") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5),
legend.position = "none")
print(p_sc)
ggsave("s24_q4.png", p_sc, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
Correlation (Price vs Quantity): 0.186
Regression: Quantity = 167.9 + 3.32 * Price (R-squared = 0.035 )
```

Output:

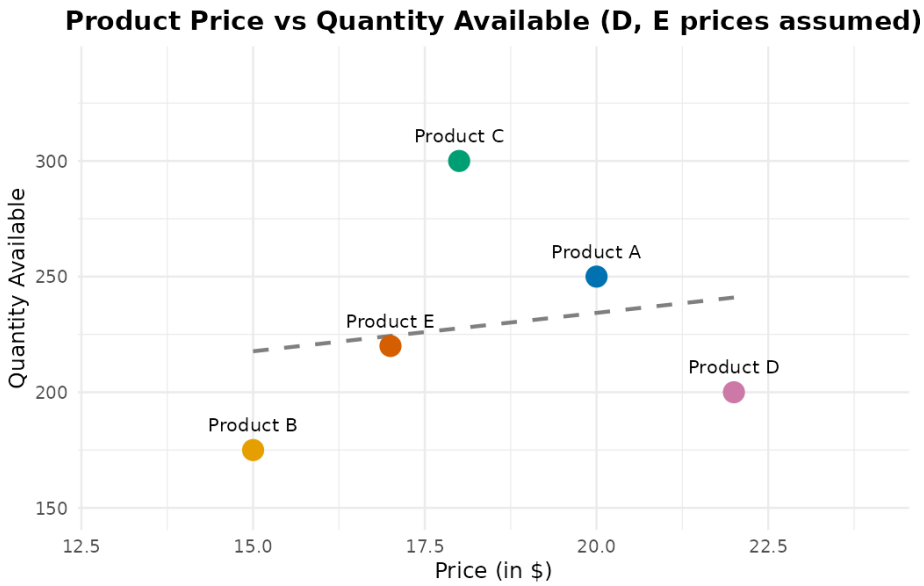


Figure 3: Price vs quantity available with a fitted line (prices partly assumed)

Inference:

There is only a **weak positive relationship** between price and quantity ($r = 0.186$, $R^2 = 0.035$). Price explains less than 4% of the variation in stock, so no useful pattern exists. Product C (300 units, $18) and Product B (175 units, $15) show that stock level is not driven by price.

This uses partly assumed prices and only five products, so the real price data should be used before drawing any conclusion.

Result

The inventory was visualised with a bar chart, a stacked bar chart and a price-quantity scatter plot in R, and a Tableau dashboard was designed. Product C has the most stock (300 units), Category 1 (assumed) holds 48% of the 1,145 units, and price has almost no relationship with quantity (r = 0.19).